Roll No. ....

71475

B. Pharma. 7th Semester (Regular/Re-Appear/Mercy Chance) Examination – December, 2025

INSTRUMENTAL METHODS OF ANALYSIS

Paper : BP701T

Time : Three Hours] *[Maximum Marks : 75*

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Section-A is ***compulsory***, each carry **2** marks, Attempt any ***two*** questions out of three from Section-**B**, each carry **10** marks and ***seven*** questions out of ***nine*** from Section-**C**, each carry **5** marks.

SECTION – A

1. Attempt ***all*** questions : $2 \times 10 = 20$

   (a) Give ***two*** applications of Electrophoresis.

   (b) Give the principle of Atomic Absorption Spectroscopy.

(c) Give differences between TLC and Paper Chromatography.

(d) Define Fermi Resonance in IR spectroscopy.

(e) Define Singlet and Triplet exited states.

(f) Give principle of Gel Chromatography.

(g) Define Isocratic and Gradient elution in HPLC.

(h) Give principle of UV-Vis spectroscopy.

(i) Give *two* limitations of Flame photometry.

(j) Define $E^{1\%}_{1cm}$ value of a compound.

SECTION – B

Note : Answer any *two* out of the following three questions :

2. Discuss the principle, instrumentation and applications of double beam UV - Vis spectrophotometer. Draw a neat and well labelled flow diagram of it. 10

3. Discuss the following :

(a) Factors affecting vibrational frequency in IR spectroscopy. 7

(b) Theory and *two* applications of affinity chromatography. 3

4. Discuss the principle, instrumentation and applications of Gas Chromatograph. Draw a neat and well labelled flow diagram of it. 10

SECTION – C

Note : Answer any ***seven*** out of the following nine questions :

5. Define Beer-Lambert's law. Discuss various causes of deviation from it. 5

6. Discuss principle and methodology of Nepheloturbidometry. 5

7. Write a note on any ***three*** detectors of HPLC unit. 5

8. Discuss principle and methodology of Thin Layer Chromatography. 5

9. Define Electrophoretic mobility. Discuss the factors affecting it. 5

10. Write a note on the burners used in flame photometry. 5

11. Discuss principle and methodology of ion exchange chromatography. 5

12. Write a note on any ***three*** detectors of IR spectrophotometer. 5

13. Discuss any ***two*** inferences observed in AAS. How they are corrected ? 5

Department of Pharmaceutical Sciences, Indira Gandhi University, Meerpur, Rewari

B. Pharm. 7th Semester, 1st Sessional Examination, Sept, 2025

Instrumental Methods of Analysis (Theory) (Paper Code- BP701T)

Date: Sept 29, 2025

Duration: 1 hr **Total marks: 30**

Question no.1 is compulsory. Draw diagram wherever required.

1. **Define the following.** **(5×2=10 marks)**
   a. Write the advantages and disadvantages of column chromatography. — TLC
   b. What is matrix interference? Give examples. — Explain the effect of ~~UV~~ solvent in UV spectroscopy.
   c. What do you mean by fluorescence?
   d. Define auxochrome and chromophore giving examples.
   e. Give the pharmaceutical application of IR Spectroscopy.

Attempt any two questions. **(2×5=10 marks)**

2. Write the principle, sources of radiation and detectors used in UV spectroscopy.
3. Discuss the types of paper chromatography with its applications.
4. What are singlet and triplet electronic states? Describe various factors affecting fluorescence.

Attempt any one question. **(1×10=10 marks)**

5. Write a note on a) Detectors used in IR b) Sample handling in IR Spectroscopy
6. Describe the methodology and applications of TLC.

Department of Pharmaceutical Sciences, IGU Meerpur,

B. Pharm. 7th Sem, 2nd Sessional, Nov, 2025

Instrumental Methods of Analysis (Paper Code- BP701T)

Date: Nov 24, 2025 Duration: 1 hr Total marks: 30

Question no.1 is compulsory. Draw diagram wherever required.

1. Define the following. (5×2=10 marks)

a. Write two applications of gel permeation chromatography.
b. Write two examples each of cation and anion exchangers
c. State the advantages and disadvantages of agarose and polyacrylamide gels.
d. Write about total combustion burner with diagram.
e. Explain the principle of flame photometry.

Attempt any two questions. (2×5=10 marks)

5. Explain in detail various factors affecting ion exchange.
6. Explain various types of interferences in AAS.
7. Explain the theoretical principle and applications of affinity chromatography.

Attempt anyone questions. (1×10=10 marks)

5. Define derivatization. Write a note on different derivatization techniques in gas chromatography and its significance.
6. Explain various types of detectors used in HPLC. Also explain the factors affecting resolution in HPLC.